SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Sarika. K S (19252103)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Sarika. K S
Signature of the student:

Application - Based Question

1) Scenario : Event Communication security

as A student organization is preparing to share confidential event details with its core members using a simple classical encryption message "MEET AT NOON"

a) Reconstruct how the encrypted message might appear if shift 4 position is applied

The scenario uses a fixed shift classical encryption technique, which the caesar cipher. In this method, every letter in the plain text is shifted by a fixed number of positions in the alphabet, Here the shift value is 4

Encryption

plain text : MEET AT NOON

using a shift of 4 position foorward

plain TEXT M E E T AT NOON

Shift +4 Q I I X E X R S S R

Example

M -> N (1)
->Q (2)
->P (3)
Q (4) M -> Q

Therefore, the encrypted message is

QIIX EX RSSR

b) Decryption by an authorized for readability

An authorized member who knows that the encryption rule uses a shift of 4 can decrypt the message by shifting every letter 4 positions backward

Ex:

Q -> M
I -> E
I -> E
X -> T
E -> A
X -> T
R -> N
S -> O
S -> O
R -> N

Therefore

Q I I X E X R S S R -> MEET AT MOON

Conclusion:-

The caesar cipher is simple and easy to Implement It provides basic confidentiality by shifting characters according to a fixed rule. However, because there are only a limited number of possible shifts, it can be easily broken using a brute-force attack. than protecting highly confidential modern.

4) Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, the file appears normal upon casual Inspection

a) Analyze how Investigators could detect the presence of hidden data within the Image

The scenario describes a logo Image that may contain hidden confidential Information. This technique is commonly associated with digital steganography, where secret Information is concealed Inside an ordinary looking file.

1. Examine the image visually:-

The Image should first be checked for unusual pixels, distorted areas, unexpected patterns or differences in colour and brightness. A suspicious Image may contain abnormalities that are not visible during normal viewing

2. check file properties and metadata:-

→File type
→File size
→Image dimensions
→creation / modification Information
→metadata
→unusual embedded Information

Unexpected metadata or an unusual file structure can indicate that the Image has been modified

3. Compare file size:-
The suspected logo can be compared with original version larger file size may indicate that additional information has embedded.

4. perform statistical analysis
Hidden Information especially when inserted into Image pixels may disturb the normal statistical characteristics of the Image.

b) Two techniques to strengthen hidden communication

Technique 1: Encrypt the secret Information before embedding
The confidential message should first be encrypted using a strong encryption algorithm. Even if an Investigator successfully extracts the hidden data appear as ciphertext and cannot be easily understood without the encryption key

Technique 2: use a robust and less detectable steganography technique

Instead of simply storing large amounts of Information in obvious pixel locations, a more robust steganography methods can distribute the hidden Information carefully across the Image. This reduces noticeable changes and make statistical detection more difficult

Investigation flow:-

Logo Image -> Metaldata/File Analysis
Inspection -> Pixel / Statistical Analysis
steganalysis Tool -> Detect / Extract Hidden data

Conclusion:-

The stronger protection the secret message should be encrypted before embedding and a robust low detectability steganographic technique should be used.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation